

UTKARSH SINGH

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Professional Summary

First-year Computer Science Engineering student specializing in AI, driven by the craft of turning ambitious ideas into working systems. I build practical AI tools across agentic workflows, backend infrastructure, and applied machine learning.

Published researcher at **ICML** on **Adaptive Termination in Agentic Multimodal RAG via Cross-Model Evidence Grounding** designing CMEGS, a claim-level grounding metric, and MIR, a principled stopping criterion that tells a retrieval agent *when it actually knows enough* a problem most systems quietly ignore.

Stack — Python · FastAPI · Node.js · Express.js · TypeScript · SQL · Pandas · NumPy · Generative AI Systems

Education

Maharaja Agrasen Institute of Technology [GGSIPU]	New Delhi, India
B.Tech in Computer Science Engineering, Specialization in Artificial Intelligence (9.3 CGPA)	Aug. 2025 – May 2029

Kamal Public Sr. Sec. School [CBSE]	New Delhi, India
Class XII PCM with Computer Science (92.4%)	April 2011 – May 2025

Skills

Programming Languages: Python, TypeScript, JavaScript, SQL, C++, C, Tkinter

AI & Machine Learning: Generative AI, Agentic AI, Multi-Agent Systems, RAG, Multimodal RAG, LLM Applications, Deep Learning, Machine Learning, PyTorch, TensorFlow, Transformers, Hugging Face

Backend & APIs: FastAPI, Flask, Node.js, Express.js, REST APIs

Frontend: React, Next.js, Tailwind CSS

Databases: PostgreSQL, MySQL, MongoDB, Redis, SQLite, Firebase, Query Optimization, Data Modeling

MLOps & DevOps: Docker, Git, GitHub Actions (CI/CD), Inference Serving, Model Deployment, API Deployment

Cloud & Deployment: AWS (S3, EC2, Lambda), Google Cloud Platform, Vercel, Render, Railway, Hugging Face Spaces

Libraries: NumPy, Pandas, scikit-learn, LangChain, Matplotlib, OpenCV

Developer Tools: VS Code, PyCharm, Jupyter, Postman, Notion

Projects

ATLASS [Adaptive Taste Learning And Suggestion System]

- Developed a fullstack recommendation engine leveraging an ML model to suggest movies and series tailored to individual user taste and viewing preferences.
- Engineered a modular frontend architecture with dedicated layers for data pipelines, model integration, UI rendering, and state management.
- Delivered an intuitive, responsive interface with dynamic filtering and interactive content discovery for an engaging end-to-end user experience.

SCOUT — AI-Powered Cross-Platform Shopping Assistant

- Built an AI shopping agent that concurrently searches seven Indian e-commerce and quick-commerce platforms (Amazon, Flipkart, Blinkit, Zepto, Instamart, Myntra, Ajio), deduplicates results, and ranks products through a six-factor weighted scoring engine.
- Architected an async FastAPI backend with an LLM-based intent router, per-session Redis-backed state, and Server-Sent Events streaming for real-time AI-generated recommendations and review summarization.

TRACTION [Telemetry-based Real-time Automotive Intelligence Platform]

- Engineered an ML-powered vehicle co-pilot for Indian mid range car drivers that translates raw sensor telemetry across 47 fields: engine, fuel, tyres, aerodynamics, cabin air quality, and electrical systems into plain language, actionable advisories.
- Built a synthetic dataset generator simulating 10,800 readings across 180 real-world Indian driving sessions, accounting for Delhi/NCR traffic patterns, seasonal AQI variation, and fuel adulteration scenarios.
- Designed a multi phase architecture spanning ML models for fuel efficiency, tyre health, and engine diagnostics, with a plain language advisory engine and Streamlit dashboard in active development.

CareFlow

- Built a full stack personal health dashboard integrating nutrition, hydration, medication reminders, BMI tracking, and real time geospatial mapping of nearby medical facilities using Leaflet.
- Engineered an Express backend with Gemini AI integration to power an intelligent health assistant capable of handling personalized queries and system guidance.
- Crafted a high performance, minimalist UI with Framer Motion and GSAP micro interactions, prioritizing data clarity and seamless user experience.

Achievements

Published Researcher at ICML (International Conference on Machine Learning) 2026

- Co-authored **Adaptive Termination in Agentic Multimodal RAG via Cross-Model Evidence Grounding**, accepted at ICML 2026, One of the most selective ML venues globally (~26% acceptance rate); published as a first-year undergraduate.

Honoured — Award of Excellence Nov. 2025

- Honored with the Academic Excellence Award, presented by Shri Suresh Jain (National Organization Secretary, Bharat Vikas Parishad) and Shri Vijay Kumar (Joint Commissioner of Police).

Finalist — ET Gen AI Hackathon May 2026

- Co-built an **Autonomous Multi-Agent AI Code Intelligence & Remediation Platform** and ranked as the Top Performing out of 3000+ candidates nationwide, competing at one of India's largest Gen AI hackathons .

Open Source Contributor — GirlScript Summer of Code & Social Summer of Code April 2026

- Selected as an open-source contributor for GirlScript Summer of Code and Social Summer of Code, contributing to community-driven software projects.